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In [23]: import pandas as pd  
import numpy as np
```

```
In [50]: #Loading data set to be reviewed and cleaned  
df = pd.read_csv('My_Projects/Urban_Forestry/Cityworks_W0/Cityworks_W0s_2025_cleaned.csv')
```

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In [51]: #Printing a structural overview of the data set  
df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4829 entries, 0 to 4828
Data columns (total 34 columns):
#   Column                Non-Null Count  Dtype
---  -
0   workorderid            4829 non-null   int64
1   description            4829 non-null   object
2   priority               4829 non-null   int64
3   status                4829 non-null   object
4   submit_to             4829 non-null   object
5   proj_start_date       4829 non-null   object
6   proj_finish_date      4829 non-null   object
7   actual_start          4829 non-null   object
8   actual_finish         4829 non-null   object
9   shop                  4829 non-null   object
10  map_page               4828 non-null   object
11  address                4827 non-null   object
12  location              4828 non-null   object
13  category               4829 non-null   object
14  date_initiated        4829 non-null   object
15  project               2714 non-null   object
16  total_cost            4829 non-null   float64
17  comments              4800 non-null   object
18  zone                  4829 non-null   object
19  request_ids           4829 non-null   object
20  resolution            3665 non-null   object
21  sr_problem_code       4829 non-null   object
22  sr_description        4829 non-null   object
23  sr_priority           4829 non-null   float64
24  projected_start_date  4829 non-null   object
25  projected_start_time  4829 non-null   object
26  projected_finish_date 4829 non-null   object
27  projected_finish_time 4829 non-null   object
28  actual_start_date     4829 non-null   object
29  actual_start_time     4829 non-null   object
30  actual_finish_date    4829 non-null   object
31  actual_finish_time    4829 non-null   object
32  initiated_date        4829 non-null   object
33  initiated_time        4829 non-null   object
dtypes: float64(2), int64(2), object(30)
memory usage: 1.3+ MB

```

```
In [52]: #Dropping unnecessary columns  
df = df.drop(columns = ['projected_start_date', 'projected_start_time', 'projected_finish_date', 'projected
```

```
In [53]: #Converting all date fields to datetime data type  
df['proj_start_date'] = pd.to_datetime(df['proj_start_date'], dayfirst=False)  
df['proj_finish_date'] = pd.to_datetime(df['proj_finish_date'], dayfirst=False)  
df['actual_start'] = pd.to_datetime(df['actual_start'], dayfirst=False)  
df['actual_finish'] = pd.to_datetime(df['actual_finish'], dayfirst=False)  
df['date_initiated'] = pd.to_datetime(df['date_initiated'], dayfirst=False)  
df
```

Out[53]:

	workorderid	description	priority	status	submit_to	proj_start_date	proj_finish_date	actual_start	actual_finish	shc
0	390348	REMOVAL BY CONTRACTOR	3	CLOSED	UF, CONTRACTOR	2025-01-02 07:44:00	2025-02-01 07:44:00	2025-01-16 12:00:00	2025-02-21 12:00:00	
1	390350	TRIM BY CONTRACTOR	3	CREW	UF, CONTRACTOR	2025-01-02 08:20:00	2025-02-01 08:20:00	2025-01-16 12:00:00	2025-02-01 08:20:00	
2	390351	TRIM BY CONTRACTOR	3	ISSUED	UF, CONTRACTOR	2025-01-02 08:58:00	2025-02-01 08:58:00	2025-01-02 12:00:00	2025-02-01 08:58:00	
3	390356	TRIM BY CONTRACTOR	3	CLOSED	UF, CONTRACTOR	2025-01-02 11:07:00	2025-01-03 11:07:00	2025-01-16 12:00:00	2025-06-18 12:00:00	
4	390359	TREE/SEEDLING PLANTING	4	CLOSED	UF, CONTRACTOR	2025-01-02 11:44:00	2026-01-02 11:44:00	2025-01-02 11:44:00	2025-01-02 12:00:00	
...	...	...	...	...	...	...	...	...	...	...
4824	404649	REMOVAL & STUMP GRINDING BY CONTRACTOR	3	ISSUED	UF, CONTRACTOR	2025-12-31 12:28:00	2026-01-30 12:28:00	2026-06-18 12:00:00	2026-01-30 12:28:00	
4825	404650	REMOVAL & STUMP GRINDING BY CONTRACTOR	3	CLOSED	UF, CONTRACTOR	2025-12-31 13:02:00	2026-01-30 13:02:00	2025-12-31 12:00:00	2026-03-12 12:00:00	

	workorderid	description	priority	status	submit_to	proj_start_date	proj_finish_date	actual_start	actual_finish	shc
4826	404651	EMERGENCY REMOVAL BY IN-HOUSE	1	CLOSED	UF, IN-HOUSE	2025-12-31 13:07:00	2026-01-01 13:07:00	2025-12-31 13:07:00	2026-01-08 12:24:00	C
4827	404652	REMOVAL & STUMP GRINDING BY CONTRACTOR	3	INPROG	UF, CONTRACTOR	2025-12-31 13:45:00	2026-01-30 13:45:00	2025-12-31 13:45:00	2026-01-30 13:45:00	
4828	407842	TREE/SEEDLING PLANTING	4	CLOSED	UF, IN-HOUSE	2025-11-21 09:03:00	2026-11-21 09:03:00	2025-11-21 09:03:00	2026-11-21 12:00:00	

4829 rows × 24 columns

```
In [54]: #Confirming data types have been changed correctly
df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4829 entries, 0 to 4828
Data columns (total 24 columns):
#   Column                Non-Null Count  Dtype
---  -
0   workorderid           4829 non-null   int64
1   description            4829 non-null   object
2   priority               4829 non-null   int64
3   status                 4829 non-null   object
4   submit_to             4829 non-null   object
5   proj_start_date        4829 non-null   datetime64[ns]
6   proj_finish_date       4829 non-null   datetime64[ns]
7   actual_start           4829 non-null   datetime64[ns]
8   actual_finish          4829 non-null   datetime64[ns]
9   shop                   4829 non-null   object
10  map_page               4828 non-null   object
11  address                 4827 non-null   object
12  location                4828 non-null   object
13  category                4829 non-null   object
14  date_initiated          4829 non-null   datetime64[ns]
15  project                 2714 non-null   object
16  total_cost              4829 non-null   float64
17  comments                4800 non-null   object
18  zone                    4829 non-null   object
19  request_ids             4829 non-null   object
20  resolution              3665 non-null   object
21  sr_problem_code         4829 non-null   object
22  sr_description          4829 non-null   object
23  sr_priority             4829 non-null   float64
dtypes: datetime64[ns](5), float64(2), int64(2), object(15)
memory usage: 905.6+ KB

```

```

In [55]: #Calculating the difference in days between 'proj_start_date' and 'proj_finish_date'
df['proj_difference'] = (df['proj_finish_date'] - df['proj_start_date']).dt.days
#Calculating the difference in days between 'actual_start' and 'actual_finish'
df['actual_difference'] = (df['actual_finish'] - df['actual_start']).dt.days
df

```

Out[55]:

	workorderid	description	priority	status	submit_to	proj_start_date	proj_finish_date	actual_start	actual_finish	shc
0	390348	REMOVAL BY CONTRACTOR	3	CLOSED	UF, CONTRACTOR	2025-01-02 07:44:00	2025-02-01 07:44:00	2025-01-16 12:00:00	2025-02-21 12:00:00	
1	390350	TRIM BY CONTRACTOR	3	CREW	UF, CONTRACTOR	2025-01-02 08:20:00	2025-02-01 08:20:00	2025-01-16 12:00:00	2025-02-01 08:20:00	
2	390351	TRIM BY CONTRACTOR	3	ISSUED	UF, CONTRACTOR	2025-01-02 08:58:00	2025-02-01 08:58:00	2025-01-02 12:00:00	2025-02-01 08:58:00	
3	390356	TRIM BY CONTRACTOR	3	CLOSED	UF, CONTRACTOR	2025-01-02 11:07:00	2025-01-03 11:07:00	2025-01-16 12:00:00	2025-06-18 12:00:00	
4	390359	TREE/SEEDLING PLANTING	4	CLOSED	UF, CONTRACTOR	2025-01-02 11:44:00	2026-01-02 11:44:00	2025-01-02 11:44:00	2025-01-02 12:00:00	
...	...	...	...	...	...	...	...	...	...	...
4824	404649	REMOVAL & STUMP GRINDING BY CONTRACTOR	3	ISSUED	UF, CONTRACTOR	2025-12-31 12:28:00	2026-01-30 12:28:00	2026-06-18 12:00:00	2026-01-30 12:28:00	
4825	404650	REMOVAL & STUMP GRINDING BY CONTRACTOR	3	CLOSED	UF, CONTRACTOR	2025-12-31 13:02:00	2026-01-30 13:02:00	2025-12-31 12:00:00	2026-03-12 12:00:00	

	workorderid	description	priority	status	submit_to	proj_start_date	proj_finish_date	actual_start	actual_finish	shc
4826	404651	EMERGENCY REMOVAL BY IN-HOUSE	1	CLOSED	UF, IN-HOUSE	2025-12-31 13:07:00	2026-01-01 13:07:00	2025-12-31 13:07:00	2026-01-08 12:24:00	C
4827	404652	REMOVAL & STUMP GRINDING BY CONTRACTOR	3	INPROG	UF, CONTRACTOR	2025-12-31 13:45:00	2026-01-30 13:45:00	2025-12-31 13:45:00	2026-01-30 13:45:00	
4828	407842	TREE/SEEDLING PLANTING	4	CLOSED	UF, IN-HOUSE	2025-11-21 09:03:00	2026-11-21 09:03:00	2025-11-21 09:03:00	2026-11-21 12:00:00	

4829 rows × 26 columns

```
In [56]: #Converting 'sr_priority' from float to integer
df['sr_priority'] = df['sr_priority'].astype(int)
df
```

Out[56]:

	workorderid	description	priority	status	submit_to	proj_start_date	proj_finish_date	actual_start	actual_finish	shc
0	390348	REMOVAL BY CONTRACTOR	3	CLOSED	UF, CONTRACTOR	2025-01-02 07:44:00	2025-02-01 07:44:00	2025-01-16 12:00:00	2025-02-21 12:00:00	
1	390350	TRIM BY CONTRACTOR	3	CREW	UF, CONTRACTOR	2025-01-02 08:20:00	2025-02-01 08:20:00	2025-01-16 12:00:00	2025-02-01 08:20:00	
2	390351	TRIM BY CONTRACTOR	3	ISSUED	UF, CONTRACTOR	2025-01-02 08:58:00	2025-02-01 08:58:00	2025-01-02 12:00:00	2025-02-01 08:58:00	
3	390356	TRIM BY CONTRACTOR	3	CLOSED	UF, CONTRACTOR	2025-01-02 11:07:00	2025-01-03 11:07:00	2025-01-16 12:00:00	2025-06-18 12:00:00	
4	390359	TREE/SEEDLING PLANTING	4	CLOSED	UF, CONTRACTOR	2025-01-02 11:44:00	2026-01-02 11:44:00	2025-01-02 11:44:00	2025-01-02 12:00:00	
...	...	...	...	...	...	...	...	...	...	...
4824	404649	REMOVAL & STUMP GRINDING BY CONTRACTOR	3	ISSUED	UF, CONTRACTOR	2025-12-31 12:28:00	2026-01-30 12:28:00	2026-06-18 12:00:00	2026-01-30 12:28:00	
4825	404650	REMOVAL & STUMP GRINDING BY CONTRACTOR	3	CLOSED	UF, CONTRACTOR	2025-12-31 13:02:00	2026-01-30 13:02:00	2025-12-31 12:00:00	2026-03-12 12:00:00	

	workorderid	description	priority	status	submit_to	proj_start_date	proj_finish_date	actual_start	actual_finish	shc
4826	404651	EMERGENCY REMOVAL BY IN-HOUSE	1	CLOSED	UF, IN-HOUSE	2025-12-31 13:07:00	2026-01-01 13:07:00	2025-12-31 13:07:00	2026-01-08 12:24:00	C
4827	404652	REMOVAL & STUMP GRINDING BY CONTRACTOR	3	INPROG	UF, CONTRACTOR	2025-12-31 13:45:00	2026-01-30 13:45:00	2025-12-31 13:45:00	2026-01-30 13:45:00	
4828	407842	TREE/SEEDLING PLANTING	4	CLOSED	UF, IN-HOUSE	2025-11-21 09:03:00	2026-11-21 09:03:00	2025-11-21 09:03:00	2026-11-21 12:00:00	

4829 rows × 26 columns

```
In [57]: #Saving cleaned data set as new csv file
df.to_csv('My_Projects/Urban_Forestry/Cityworks_W0/Cityworks_W0s_2025_cleaned.csv', index=False)
```

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In [ ]:
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